

Notes on Approximation Theory

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Some notes that may be useful when finding approximation error bounds that are explicit, with no hidden constants and without introducing transcendental or trigonometric functions.

The notes generally relate to error bounds on how close a polynomial is to a single-variable function on a closed interval. The mapping from a function to a function (in this case, from a single-variable function to a polynomial “close” to it) is called an *operator*, and operators involved in these error bounds are often linear operators, whose behavior is relatively simple to examine.

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2 Notation and Definitions

For definitions of *continuous*, *derivative*, *convex*, *concave*, *Hölder continuous*, and *Lipschitz continuous*, see the definitions section in “**Supplemental Notes for Bernoulli Factory Algorithms**¹”.

- The *closed unit interval* (written as $[0, 1]$) means the set consisting of 0, 1, and every real number in between.

¹<https://peteroupc.github.io/bernsupp.html#Definitions>

- An *operator* is a mapping from a function to a function.
- An operator L is *linear* if it satisfies $L(af) = aL(f)$ and $L(f + g) = L(f) + L(g)$ for all input functions f and g and every number a .
- An operator L is *positive* if it has the property that, if an input function f is nonnegative on its domain, so is $L(f)$.²
- The *operator norm* of an operator L is the maximum absolute value of $L(f)$ over all input functions f with a maximum absolute value 1 or less. This assumes L maps continuous functions on a closed interval to functions of that kind.
- In this document, e_i is a function such that $e_i(t) = t^i$, so that $e_0(t) = 1$ and $e_1(t) = t$; as an example, if $L(f) = f(0) + f(1)$, then $L(e_1 - x) = (e_1(0) - x) + (e_1(1) - x) = (0 - x) + (1 - x) = 1 - 2x$.
- The *expected value* (or mean or “long-run average”) of a random variable Y is denoted $\mathbb{E}[Y]$.
- A *modulus of continuity of order 1* of a function f , denoted $\omega_1(\delta)$, means a nonnegative and nowhere decreasing function where, for each $\delta \geq 0$, $\text{abs}(f(x) - f(y)) \leq \omega_1(\delta)$ whenever x and y are in f ’s domain and no more than δ apart. Loosely speaking, $\omega_1(\delta)$ gives how much f can vary when f is restricted to a window of size δ or less. The modulus of continuity reflects the “regularity” of f ; generally, the smaller it is, the more “regular”.

3 Bernstein Form and Bernstein Polynomials

Among the best known examples of linear operators are the Bernstein polynomials.

In this document, a polynomial $P(x)$ is written in *Bernstein form of degree n* if it is written as—

$$P(x) = \sum_{k=0}^n a_k \frac{n!}{(k!)((n-k)!)} x^k (1-x)^{n-k},$$

where the real numbers $a_0, \dots, a_n$ are the polynomial’s *Bernstein coefficients*.³

The degree- n *Bernstein polynomial* of an arbitrary function $f(x)$ has Bernstein coefficients $a_k = f(k/n)$. In general, this Bernstein polynomial differs from f even if f is a polynomial. In this document, the degree- n Bernstein polynomial of f is denoted $B_n(f)$. $B_n(f)$ is a positive linear operator.

4 “Moments” of Linear Operators

To examine the approximation behavior of linear operators, it is helpful to find the so-called “moments” of those operators, that is, the functions they map certain functions to.

For a linear operator L , they are:

- “Raw moments”: The values of $L(e_i)$ for each integer $i \geq 0$.
- “Central moments”: The values of $L((e_1 - x)^i)$ for each integer $i \geq 0$. If the “raw moments” $L(e_0), \dots, L(e_j)$ are known, then $L((e_1 - x)^j)$ is also known, thanks to proposition 5.6 of Gonska et al. (2006)⁴.

²A better term for positive operators is probably nonnegativity-preserving operators.

³ $n! = 1*2*3*\dots*n$ is also known as n factorial; in this document, $(0!) = 1$. *Summation notation*, involving the Greek capital sigma (Σ), is a way to write the sum of one or more terms of similar form. For example, $\sum_{k=0}^n g(k)$ means $g(0) + g(1) + \dots + g(n)$, and $\sum_{k \geq 0} g(k)$ means $g(0) + g(1) + \dots$.

⁴Gonska, Heiner, Paula Pi ul, and Ioan Raşa. “On differences of positive linear operators.” *Carpathian Journal of Mathematics* (2006): 65-78.

- “Absolute moments”: The values of $L(\text{abs}(e_1 - x)^i)(x)$ for each integer $i \geq 0$. When i is even, $L(\text{abs}(e_1 - x)^i) = L((e_1 - x)^i)$.

Because L is linear, if $L(e_i) = e_i$ for each i from 0 through j (j is zero or a positive integer), then:

- L reproduces all polynomials up to degree j (that is, $L(f) = f$ whenever f is a polynomial of degree j or less).
- The 0-th “central moment” is $L(e_0) = L((e_1 - x)^0) = 1$, and for each i from 1 through j , $L((e_1 - x)^i) = 0$.

Also, because L is linear, the “moments” of degree up to m , say, lead to easy ways to find the mapping by L of any polynomial of degree up to m , when the polynomial is written in “power” form.

Example: Let $f(x)$ be the polynomial $4x^3 - 6x^2 + 8x^1 - 10$. Then:

$$L(f) = 4L(e_3) - 6L(e_2) + 8L(e_1) - 10L(e_0).$$

4.1 “Moments” of Bernstein Polynomials

The following results deal with useful quantities when discussing the error in approximating a function by Bernstein polynomials.

Suppose a coin shows heads with probability p , and n independent tosses of the coin are made, where n is 1 or greater. Then the total number of heads X follows a *binomial distribution*. The following are useful quantities of this distribution.

- $T_{n,r}$: The *central moment* (moment about the mean) of X is denoted $T_{n,r}(p) = \mathbb{E}[(X - \mathbb{E}[X])^r] = B_n((e_1 - p)^r)(p) \cdot n^r$. Formulas for computing this central moment are given in Skorski (2024)⁵.
- $S_{n,r}$: Traditionally, the central moment of X/n or the ratio of heads to tosses is denoted $S_{n,r}(p) = T_{n,r}(p)/n^r = \mathbb{E}[(X/n - \mathbb{E}[X/n])^r] = B_n((e_1 - p)^r)(p)$. (T and S are notations of S.N. Bernstein, known for Bernstein polynomials.) $S_{n,r}$ is thus the r -th “central moment” of degree- n Bernstein polynomials.
- $M_{n,r}$: The r -th *central absolute moment* of X/n is denoted $M_{n,r}(p) = \mathbb{E}[\text{abs}(X/n - \mathbb{E}[X/n])^r] = B_n(\text{abs}(e_1 - p)^r)(p)$. If r is even, $M_{n,r}(p) = S_{n,r}(p)$. $M_{n,r}$ is thus the r -th “absolute moment” of degree- n Bernstein polynomials.

The following gives bounds on $M_{n,r}$; some results in approximation theory rely on bounds like these.⁶

Proposition 1: Let $r \geq 0$, and let $\sigma(r, t) = (r!)/(((r/2)!)t^{r/2})$. Then for real numbers r and integers n described in the following table, $M_{n,r}(p) \leq \mu_{n,r}/n^{r/2}$, where $\mu_{n,r}$ is as given in the table.

If r ...	Then $\mu_{n,r}$ is...
Is an even integer.	$\sigma(r, 6)$, for positive n .
Is an even integer, but not greater than 44.	$\sigma(r, 8)$, for positive n .
Is 1.	$1/2$, for positive n .
Is odd, and $3 \leq r \leq 43$.	$\sqrt{\sigma(r-1, 8)\sigma(r+1, 8)} = r^{1/2}(r-1)!/(2 \cdot 8^{(r-1)/2}((r-1)/2)!)$, for $n \geq 2$.

⁵Skorski, Maciej. “Handy formulas for binomial moments.” *Modern Stochastics: Theory and Applications* 12.1 (2024): 27-41.

⁶It is also possible to bound the “absolute moment” as $M_{n,r}(p) \leq C(r)(\max(1/n, (p(1-p)/n)^{1/2}))^r$ or $M_{n,r}(p) \leq D(r)(1/n + (p(1-p)/n)^{1/2})^r$ (G.G. Lorentz, “The degree of approximation by polynomials with positive coefficients”, 1966), but the constants $C(r)$ and $D(r)$ seem to be higher (and less favorable) than the $E(r)$ in $M_{n,r}(p) \leq E(r)/n^{r/2}$.

If $r...$	Then $\mu_{n,r}$ is...
Is odd and greater than 43.	$\sqrt{\sigma(r-1, 6)\sigma(r+1, 6)}$, for $n \geq 2$.

Proof: The first row comes from a result of Adell and Cárdenas-Morales (2018)⁷. The second row is an improved result of the first, from Molteni (2022)⁸. The third row follows from Cheng (1983)⁹. The fourth and fifth rows follow from the first and second as well as that the absolute central moment for odd r can be bounded for every integer $n \geq 2$, using Schwarz’s inequality (Weisstein)¹⁰ (see also Bojanić and Shisha 1975¹¹ for the case $r = 4$). []

5 Taylor Expansion of Linear Operators

Continuous functions can be “unwrapped” into a Taylor expansion. The linear mapping of those functions also has a Taylor expansion of sorts, which is described next.

Let $f(\lambda)$ have a continuous s -th derivative on a closed interval, where s is zero or a positive integer, and let $L(f)$ be a linear operator that maps continuous functions on that interval to functions of that kind. Then:

$$L(f)(\lambda) = L(R_s(f, \lambda)) + \sum_{i=0}^s L((e_1 - \lambda)^i)(\lambda) \frac{f^{(i)}(\lambda)}{i!}, \quad (1)$$

where $R_s(f, \lambda)$ is the remainder after subtracting from f the degree- s Taylor polynomial of f centered at λ . (See also Pițul (2007, proof of theorem 5.8)¹².) $R_s(f, \lambda)$ is 0 if f is a polynomial of degree s or less.

If L reproduces constants, so that $L(e_0) = 1$, this becomes:

$$L(f)(\lambda) - f(\lambda) = L(R_s(f, \lambda)) + \sum_{i=1}^s L((e_1 - \lambda)^i)(\lambda) \frac{f^{(i)}(\lambda)}{i!}. \quad (2)$$

If L reproduces polynomials up to degree s , this even reduces to $L(f)(\lambda) - f(\lambda) = L(R_s(f, \lambda))$.

It can be seen from the expansions just given that finding upper bounds for $L_n(f)(\lambda)$ involves:

- Finding upper bounds for L ’s “central moments” up to the s -th order.
- Finding upper bounds for $L(R_s(f, \lambda))$.
 - If L is positive linear, such bounds are given in the section “**Bounds for General Positive Linear Operators**”.

⁷Adell, J.A., Cárdenas-Morales, D., “**Quantitative generalized Voronovskaja’s formulae for Bernstein polynomials**”, Journal of Approximation Theory 231, July 2018. <https://www.sciencedirect.com/science/article/pii/S0021904518300376>

⁸Molteni, Giuseppe. “Explicit bounds for even moments of Bernstein’s polynomials.” Journal of Approximation Theory 273 (2022): 105658.

⁹Cheng, F., “On the rate of convergence of Bernstein polynomials of functions of bounded variation”, Journal of Approximation Theory 39 (1983).

¹⁰Weisstein, Eric W. “Schwarz’s Inequality.” From MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/SchwarzsInequality.html>

¹¹R. Bojanic, O. Shisha, “Degree of L^1 approximation to integrable functions by modified Bernstein polynomials”, Journal of Approximation Theory 13, 66–72 (1975).

¹²Pițul, P., “Evaluation of the Approximation Order by Positive Linear Operators”, dissertation, Universität Duisberg-Essen, 2007.

- If L is not positive, finding such bounds is harder. This situation can be helped if L can be written as a difference between two positive linear operators LA and LB , so that $L(f) = LA(f) - LB(f)$.¹³ See the “**Example**” section later in this document.
- If L is linear and reproduces all polynomials up to degree s , and f has a continuous $(s + 1)$ -th derivative, the so-called *Peano kernel theorem* can help, but it’s difficult to apply generically to such an operator.¹⁴ See (Waldron 1999)¹⁵; see also (Brass and Förster 1998)¹⁶.

Meanwhile, bounds for the derivatives of f (here, $f^{(i)}$) are often assumed to be known beforehand.

6 Results on Error Bounds

Some results on error bounds for certain classes of operators.

6.1 Bounds for General Positive Linear Operators

The following results give bounds that apply to large classes of positive linear operators. In this section:

- $\sigma_i = L((e_i - \lambda)^i)(\lambda)$ (the i -th “central moment” of the linear operator L in question).
- $\tau_i = L(\text{abs}(e_i - \lambda)^i)(\lambda)$ (the i -th “absolute moment” of the linear operator L in question).
- $\omega_1(f, \delta)$ is the smallest modulus of continuity of a function f of order 1, with parameter δ .
- $\tilde{\omega}_1(f, \delta)$ is the smallest concave modulus of continuity of f of order 1, both with parameter δ .

Lemma 1. *Let $f(\lambda)$ be continuous on a closed interval, and let L be a positive linear operator that maps continuous functions on that interval to functions of that kind and reproduces all constants (so that $L(e_0) = 1$). Then $\text{abs}(L(f)(\lambda) - f(\lambda)) \leq \tilde{\omega}_1(f, \tau_1)$.*

Proof: Follows from a result of Gonska and Meier (1985, theorem 3.1)¹⁷. []

Lemma 2. *Let $f(\lambda)$ be continuous on a closed interval, and let L be a positive linear operator that maps continuous functions on that interval to functions of that kind and reproduces all polynomials up to degree 1 (constants and linear functions). Let $h > 0$ be a real number. Then:*

No.	If f ...	Then $\text{abs}(L(f)(\cdot) - f(\cdot)) \dots$
1	Is continuous.	$(1 + (\sigma_2)^{1/2}/h)\omega_1(f, h)$.
2	Is continuous.	$(1 + (\sigma_2)/h^2)\omega_1(f, h)$.
3	Is continuous.	(Use ineq. 1 if $h < (\sigma_2)^{1/2}$, or ineq. 2 otherwise.)
4	Is continuous.	$\tilde{\omega}_1(f, (\sigma_2)^{1/2})$.

¹³I suspect that, whenever L is a bounded linear operator that maps continuous functions on a closed interval to functions of that kind, L can be written as a difference between two positive linear operators. But I have not seen a proof of that statement; Acu et al. (“**Grüss-type and Ostrowski-type inequalities in approximation theory**”, Ukr Math J 63, 843–864, 2011) give a similar statement but without proof. <https://doi.org/10.1007/s11253-011-0548-2>

¹⁴For example, if L is linear but not positive, applying the theorem may require calculating mappings of truncated power functions, which are generally not polynomials (for example $L(\max(0, e_0 - x)^i)$ for some $i > 0$).

¹⁵Waldron, Shayne. “Refinements of the Peano kernel theorem.” Numerical functional analysis and optimization 20.1-2 (1999): 147-161.

¹⁶Brass, Helmut, and Klaus-Jürgen Förster. “On the application of the Peano representation of linear functionals in numerical analysis.” Recent progress in inequalities. Dordrecht: Springer Netherlands, 1998. 175-202.

¹⁷Gonska, H.H., Meier, J., “On approximation by Bernstein-type operators: best constants”, Studia Sci. Math. Hungar. 22, 1987.

No.	If $f \dots$	Then $\$ \text{abs}(L(f)(\cdot) - f(\cdot)) \dots \$$
5	Is Hölder continuous with Hölder exponent α ($0 < \alpha \leq 1$) and Hölder constant M or less.	$M(\sigma_2)^{\alpha/2}$.
6	Is Lipschitz continuous with Lipschitz constant M or less.	$M(\sigma_2)^{1/2}$.
7	Has a continuous derivative.	$((h+2)^2/(8h)) \cdot \omega_1(f^{(1)}, h \cdot \sqrt{\sigma_2}) \cdot \sqrt{\sigma_2}$.
8	Has a continuous derivative.	$\frac{1}{2}(\sigma_2)^{1/2} \tilde{\omega}_1(f^{(1)}, (\sigma_2)^{1/2})$.
9	Has a Hölder-continuous derivative with Hölder exponent α ($0 < \alpha \leq 1$) and Hölder constant M or less.	$\frac{M}{2}(\sigma_2)^{(1+\alpha)/2}$.
10	Has a Lipschitz-continuous derivative with Lipschitz constant M or less.	$\frac{M}{2}(\sigma_2)$.

Proof: Inequality 1 follows from a special case of a theorem on positive linear operators from Shisha and Mond (1968)¹⁸; inequality 2 follows from a result of Mond (1978)¹⁹; inequality 3, a result of Păltănea (2004, corollary 1.2.2)²⁰; and inequality 4, a result of Peetre (1969)²¹. Inequality 8 follows from a result of Gonska and Meier (1985, theorem 4.1)²²; see also Păltănea and Dimitriu (2016, remark 3)²³. Inequality 7 is a special case of Theorem 2.19 (in conjunction with Remark 2.21) of Anastassiou (1985), with the interval $[a, b]$, $m = 1$ (since the function is defined on all of $[a, b]$), $r = h$, and x_0 equal to λ .

Inequality 5 follows from inequality 4 using properties of Hölder-continuous functions; by assumption, f admits the continuous and concave modulus of continuity $\omega_1(\delta) = M\delta^\alpha$, where $\delta > 0$. Inequality 9 follows from inequality 8 in the same manner. Inequalities 6 and 10 follow from inequalities 5 and 9, respectively, given that Lipschitz-continuous functions are Hölder continuous with Hölder exponent 1. []

Lemma 3. *Let $f(\lambda)$ have a continuous k -th derivative on a closed interval, and let L be a positive linear operator that maps continuous functions on that interval to functions of that kind. Let $h > 0$ be a real number. Then $L(f)(\lambda) = L(Q_k(f, \lambda))(\lambda) + L(R_k(f, \lambda))(\lambda)$, where:*

$$\text{abs}(L(Q_k(f, \lambda)) - f(\lambda)) \leq \left(\sum_{i=0}^k \frac{\max(\text{abs}(f^{(i)})) \text{abs}(\sigma_i)}{i!} \right),$$

$$\text{abs}(L(R_k(f, \lambda))) \leq \left(\frac{\tau_k}{k!} + \frac{\tau_{k+1}}{(k+1)! \cdot h} \right) \cdot \omega_1(f^{(k)}, h),$$

$$\text{and } \text{abs}(L(R_k(f, \lambda))) \leq \max \left(\frac{\tau_k}{k!}, \frac{\tau_{k+1}}{(k+1)! \cdot 2h} \right) \cdot \tilde{\omega}_1(f^{(k)}, 2h),$$

$$\text{and } \text{abs}(L(R_k(f, \lambda))) \leq \frac{\tau_k}{k!} \cdot \tilde{\omega}_1(f^{(k)}, \frac{\tau_{k+1}}{(k+1)\tau_k}),$$

and where:

¹⁸Shisha, O., Mond, B., “The degree of convergence of linear positive operators”, 1968.

¹⁹Mond, B., “On the degree of approximation by linear positive operators”, *Journal of Approximation Theory* 18 (1976).

²⁰Păltănea, R., *Approximation Theory Using Positive Linear Operators*, Birkhäuser, 2004.

²¹Peetre, J., “On the connection between the theory of interpolation spaces and approximation theory”, in *Approximation Theory*, 1969.

²²Gonska, H.H., Meier, J., “On approximation by Bernstein-type operators: best constants”, *Studia Sci. Math. Hungar.* 22, 1987.

²³Păltănea, R., Dimitriu, M.T., “On some second order moduli of smoothness.” *General Mathematics* 24 (2016)

- $Q_k(f, \lambda) = \sum_{i=0}^k f^{(i)}(\lambda) \cdot (e_0 - \lambda)^i / (i!)$ is the degree- k Taylor polynomial of f centered at λ .
- $R_k(f, \lambda)$ is the Taylor remainder that results from subtracting $Q(f, \lambda)$ from f .

Proof: The second to fourth bounds given relate to the Taylor remainder. The second bound comes from Păltănea and Smuc (2019, Theorem 1)²⁴; the third bound comes from corollary 3.2 of Dimitriu (2010)²⁵ and Brudnyi's lemma; and the fourth bound follows from the second with $h = \tau_{k+1}/(2(k+1)\tau_k)$ and comes from Gonska et al. (2006)²⁶, where the closed interval assumed was the closed unit interval; see also Gonska (2007)²⁷, Pițul (2007)²⁸. See also Anastassiou (1985, theorem 2.31)^{29,30} []

Lemma 4. *Let k be zero or a positive integer. Let $f(\lambda)$ —*

1. *have a Lipschitz-continuous k -th derivative on a closed interval, with Lipschitz constant M or less, or*
2. *have a continuous $(k+1)$ -th derivative on that interval, with maximum absolute value M or less,*

and let L be a positive linear operator that maps continuous functions on that interval to functions of that kind. Then $\text{abs}(L(R_k(f, \lambda))) \leq M\tau_{k+1}/((k+1)!)$, where $R_k(f, \lambda)$ is as in Lemma 3.

Proof: Follows from the third bound for $L(R_k(f, \lambda))$ in Lemma 3 in the same manner as inequality 10 of Lemma 2, assuming f has property 1 listed on the present lemma. This lemma remains true if f has property 2 instead of property 1 because, by the so-called Hardy–Littlewood lemma, every function with property 2 has property 1 (but not vice versa). []

The following lemma adapts the previous lemmas to the setting of random variables.

Lemma 5. *Let $f(\lambda)$ be continuous on a closed interval, and let Y be a random variable taking only values in that interval. Then Lemmas 1 through 4 apply as appropriate to f meeting their conditions, with $L(f) = \mathbb{E}[f(Y)]$ and $\lambda = \mathbb{E}[Y]$.*

Proof: With these assumptions there is a positive linear operator $L(f) = \mathbb{E}[f(Y)]$ for Y and f , according to Theorem 3.1.1 of Frantz (1984)³¹, letting $x_o = \lambda$. Then $L(e_0) = \mathbb{E}[e_0(Y)] = \mathbb{E}[1] = 1$ regardless of Y , and $L(e_1) = \mathbb{E}[e_1(Y)] = \mathbb{E}[Y] = \lambda$, so L reproduces all polynomials of degree up to 1. []

6.2 Bounds for Remainder of Bernstein Polynomials

The following results specialize the previous ones to the case of **Bernstein polynomials** B_n . They apply to the Bernstein polynomial of the result of subtracting a Taylor polynomial from a function, and are useful when a linear operator contains $B_n(f)$ in its definition and reproduces all polynomials of degree r or less.

Lemma 6: *Let k be zero or a positive integer. Let $f(\lambda)$ —*

²⁴Păltănea, R., Smuc, M., “Sharp Estimates of Asymptotic Error of Approximation by General Positive Linear Operators in Terms of the First and the Second Moduli of Continuity”, *Results in Mathematics* 74 (2019).

²⁵Dimitriu, M.T., “**Estimates with optimal constants using Peetre’s K-functionals**”, *Carpathian Journal of Mathematics* 26 (2010). <https://www.jstor.org/stable/43964559>

²⁶Gonska, Heiner, Paula Pițul, and Ioan Raşa. “On Peano’s form of the Taylor remainder, Voronovskaja’s theorem and the commutator of positive linear operators”, 2006.

²⁷Gonska, Heiner. “On the degree of approximation in Voronovskaja’s theorem”, *Studia Univ. Babeş-Bolyai, Math.*, September 2007.

²⁸Pițul, P., “Evaluation of the Approximation Order by Positive Linear Operators”, dissertation, Universität Duisburg-Essen, 2007.

²⁹Anastassiou, George A. “**A study of positive linear operators by the method of moments, one-dimensional case.**” *Journal of Approximation Theory* 45.3 (1985): 247-270. <https://www.sciencedirect.com/science/article/pii/0021904585900498>

³⁰The paper Cichoń et al., “**On delta-method of moments and probabilistic sums**”, *ANALCO* 2013, has very similar results, but they assume the function f has a k -th derivative defined on an *open* interval (say, $0 < \lambda < 1$), rather than a *closed* one, making those results harder to use if Y is a random variable that can take a value equal to either endpoint of the interval (in this example, 0 or 1). <https://doi.org/10.1137/1.9781611973037.11>

³¹Frantz, Deborah A. *Summability methods, probability distributions, and associated positive linear operators*. Lehigh University, 1984.

1. have a Lipschitz-continuous k -th derivative on the closed unit interval, with Lipschitz constant M or less, or
2. have a continuous $(k+1)$ -th derivative on that interval, with maximum absolute value M or less.

Then the following bound holds true: $\text{abs}(B_n(R_k(f, \lambda))) \leq (M\mu_{k+1})/((k+1)!n^{(k+1)/2})$ for every integer $n \geq 2$ (and also for $n = 1$ if k is odd), where μ_k is as defined in Proposition 1.

Proof: Follows from Lemma 4, with $L(f) = B_n(f)$, and from Proposition 1. []

Corollary 1: Let $f(\lambda)$, k , and M be as in Lemma 6. Then, for every $0 \leq \lambda \leq 1$:

If k is:	Then $\text{abs}(B_n(R_k(f, \lambda))) \leq \dots$
0.	$M(1/2)/n^{1/2}$ for every integer $n \geq 1$.
1.	$M(1/8)/n = 0.125M/n$ for every integer $n \geq 1$.
2.	$M(\sqrt{3}/48)/n^{3/2} < 0.3609M/n^{3/2}$ for every integer $n \geq 2$.
3.	$M(1/128)/n^2 = 0.0078125M/n^2$ for every integer $n \geq 1$.
4.	$M(\sqrt{5}/1280)/n^{5/2} < 0.001747/n^{5/2}$ for every integer $n \geq 2$.
5.	$M(1/3072)/n^3 < 0.0003256/n^3$ for every integer $n \geq 1$.

6.3 Whitney's Inequality on Polynomial Errors

The following inequality gives a bound on the “best possible” error that a polynomial of degree n can achieve in approximating a function.

Let n be zero or a positive integer, let $f(\lambda)$ be continuous on a closed interval $[a, b]$, and let P be a polynomial of degree n or less with the least maximum absolute difference between f and the polynomial on that interval. Then the error of P in approximating f is bounded as follows (see Babenko and Kryakin 2019³²):

$$\|f - P\|_\infty \leq W \cdot \omega_{n+1}(f, \frac{b-a}{n+1}),$$

where—

- W is:
 - 1 if $n \leq 7$.
 - $(2 + \exp(-2))(< 2.13534)$ if $n \geq 8$.
 - $3/4$ if $n = 1$ and f is convex (Singh Kaire and Prymak 2023/2025)³³.
 - $1/2$ if $n = 1$, f is convex, and $a = -b$ (Singh Kaire and Prymak 2023/2025)³⁴.
- $\|g\|_\infty$ is the maximum of the absolute value of (the continuous function) g on $[a, b]$, and
- $\omega_n(f, h)$ is the smallest modulus of continuity of f of order n , with parameter h .

Using properties of moduli of continuity (see Sevy 1991³⁵, sec. 2.0.2; Gonska 1985³⁶), if f has a continuous $(n+1)$ -th derivative on $[a, b]$:

³²Babenko, Alexander G., and Yuriy V. Kryakin. “Special difference operators and the constants in the classical Jackson-type theorems.” Topics in Classical and Modern Analysis: In Memory of Yingkang Hu. Cham: Springer International Publishing, 2019. 35-46.

³³Jaskaran Singh Kaire and Andriy Prymak. “Whitney-type estimates for convex functions.” arXiv preprint arXiv:2311.00912 (2023).

³⁴Jaskaran Singh Kaire and Andriy Prymak. “Whitney-type estimates for convex functions.” arXiv preprint arXiv:2311.00912 (2023).

³⁵Sevy, J., “Acceleration of convergence of sequences of simultaneous approximants”, dissertation, Drexel University, 1991.

³⁶H. H. Gonska, *Quantitative Approximation in $C(X)$* , Habilitationsschrift, Universität Duisburg, 1985.

$$\|f - P\|_\infty \leq W \cdot \left(\frac{b-a}{n+1}\right)^{n+1} \|f^{(n+1)}\|_\infty,$$

and if f has a continuous n -th derivative on that interval:

$$\|f - P\|_\infty \leq W \cdot \left(\frac{b-a}{n+1}\right)^n \omega_1(f^{(n)}, \frac{b-a}{n+1}).$$

6.4 Another Inequality on Polynomial Errors

Like Whitney's inequality, the following gives a bound on the “best possible” error between a polynomial and a function.

Let n be zero or a positive integer, let $f(\lambda)$ have a continuous $(n+1)$ -th derivative on the closed interval $[-1, 1]$, «It would be interesting to find a version of this inequality that works for any closed interval $[a, b]$.» and let P be a polynomial of degree n or less with the least maximum absolute difference between f and the polynomial on that interval. Then the error of P in approximating f is bounded as follows (Phillips 2003, theorem 2.4.6)³⁷:

$$\|f - P\|_\infty \leq \frac{1}{2^n} \frac{\|f^{(n+1)}\|_\infty}{(n+1)!}.$$

6.5 Lebesgue Inequality for Certain Linear Operators

Let $f(\lambda)$ be a continuous function on a closed interval. For any sequence of linear operators (L_n) that map continuous functions to polynomials and reproduce all polynomials up to degree $m(n)$ (which depends on n), the following error bound (also known as *Lebesgue's lemma* or the *Lebesgue inequality*) holds true for each n :

$$\text{abs}(L_n(f)(x) - f(x)) \leq (1 + \|L_n\|) \cdot \max_t (\text{abs}(f(t) - P(t))),$$

where $\|L_n\|$ is the operator norm of L_n , and P is a polynomial of degree up to $m(n)$ with the least maximum absolute difference between f and the polynomial (see also DeVore and Lorentz (1993)³⁸, Cheney (1996, chapter 6)³⁹). But this error bound will generally be crude or trivial unless L_n are nonpositive operators. Indeed, the only positive linear operator L that reproduces all polynomials up to degree 2 is the identity operator $L = f$.⁴⁰

Example: Let f have a continuous third derivative on the closed unit interval. Combining the previous inequality with the Whitney-type inequalities given earlier leads to the following error bound for linear operators L that map continuous functions to polynomials and reproduce all polynomials up to degree 2:

³⁷Phillips, G.M., *Interpolation and Approximation by Polynomials*, Springer, 2003. <https://doi.org/10.1007/b97417>

³⁸R.A. DeVore and G.G. Lorentz, *Constructive Approximation*, 1993.

³⁹E. W. Cheney, *Introduction to Approximation Theory*, 1998.

⁴⁰Guessab, A., Nouisser, O. & Schmeisser, G. Enhancement of the algebraic precision of a linear operator and consequences under positivity. *Positivity* 13, 693–707 (2009). <https://doi.org/10.1007/s11117-008-2253-4>. However, Gavrea and Ivan (“A note on the fixed points of positive linear operators”, *Journal of Approximation Theory* (227), 2018) pointed out that there are positive linear operators besides the identity that reproduce all polynomials of the form x^i where $i > 0$. <https://doi.org/10.1016/j.jat.2017.12.001>

$$\begin{aligned}
\text{abs}(L(f)(x) - f(x)) &\leq (1 + \|L\|) \cdot 1 \cdot \left(\frac{1}{3}\right)^3 \|f^{(3)}\|_\infty \\
&= (1 + \|L\|) \|f^{(3)}\|_\infty / 27.
\end{aligned}$$

7 Example

This example shows how to find a linear operator's bounds.

Let $L_n(f)$ be a linear operator inspired by **a conjecture I have**⁴¹ on polynomial approximation. It is described as follows:

$$\begin{aligned}
L_n(f)(\lambda) &= \sum_{i=0}^n \left(W_{2n}(f) \left(\frac{k}{2n} \right) - W_n(f) \left(\frac{i}{n} \right) \right) \sigma_{n,k,i} \\
&= \mathbb{E} \left[W_{2n}(f) \left(\frac{k}{2n} \right) - W_n(f) \left(\frac{X_k}{n} \right) \right],
\end{aligned}$$

where:

- $k = 2n\lambda$, where $0 \leq \lambda \leq 1$.
- $W_n(f)$ is a linear operator that approaches f as n increases.⁴²
- X_k is a hypergeometric($2n, k, n$) random variable.
- $\sigma_{n,k,i}$ equals $\binom{n}{i} \binom{n}{k-i} / \binom{2n}{k}$ and is the probability that X_k equals i .
- $\mathbb{E}[Y]$ is the expected value (or mean or “long-run average”) of the random variable Y .

L_n and W_n are generally nonpositive operators. As an example, take $W_n = 2f - B_n(f)$. Then $B_n(W_n(f))$ is a linear operator that is the iterated Boolean sum of degree- n Bernstein polynomials, with one iteration; see Güntürk and Li (2021a, Theorem 5)⁴³. That paper, among others (for example, Micchelli 1973⁴⁴), showed that this operator approaches f at the rate $O(1/n^{3/2})$ if f has a continuous third derivative. (“ $O(1/n^{3/2})$ ” means the error is no greater than a constant times $1/n^{3/2}$ for all values of n .)

With this choice of W_n , L_n becomes:

$$\begin{aligned}
L_n(f)(\lambda) &= \sum_{i=0}^n \left((2f \left(\frac{k}{2n} \right) - B_{2n}(f) \left(\frac{k}{2n} \right)) - (2f \left(\frac{i}{n} \right) - B_n(f) \left(\frac{i}{n} \right)) \right) \sigma_{n,k,i} \\
&= \mathbb{E} \left[(2f \left(\frac{k}{2n} \right) - B_{2n}(f) \left(\frac{k}{2n} \right)) - (2f \left(\frac{X_k}{n} \right) - B_n(f) \left(\frac{X_k}{n} \right)) \right] \\
&= \sum_{i=0}^n \left((2f \left(\frac{k}{2n} \right) + B_n(f) \left(\frac{i}{n} \right)) \right) \sigma_{n,k,i} - \sum_{i=0}^n \left((2f \left(\frac{i}{n} \right) + B_{2n}(f) \left(\frac{k}{2n} \right)) \right) \sigma_{n,k,i}
\end{aligned}$$

⁴¹https://peteroupc.github.io/bernsupp.html#A_Conjecture_on_Polynomial_Approximation

⁴² W_n can, in principle, be nonlinear instead, but this would require a totally different approach to finding the approximation error, and L_n would then be nonlinear in general.

⁴³Güntürk, C. Sinan, and Weilin Li. “**Approximation with one-bit polynomials in Bernstein form**”, arXiv:2112.09183 (2021); Constructive Approximation, pp.1-30 (2022). <https://arxiv.org/pdf/2112.09183>

⁴⁴Micchelli, Charles. “**The saturation class and iterates of the Bernstein polynomials**”, Journal of Approximation Theory 8, no. 1 (1973): 1-18. <https://www.sciencedirect.com/science/article/pii/0021904573900282>

$$= LA_n(f)(\lambda) - LB_n(f)(\lambda).$$

Here, LA_n and LB_n are positive linear operators, making it easier to assess their approximation properties.

It will be shown that, if f has a continuous third derivative, the rate of L_n towards zero is $O(M/n^{3/2})$, where M is the maximum absolute value of f and its derivatives up to the third derivative. The proof of this relies on exact expressions of L_n 's “**raw moments**” and “**central moments**”, and those for the combined operator $(LA_n + LB_n)$.

The following are some of these values and those for related operators:

- $L_n(e_0)(x) = L_n((e_1 - x)^0)(x) = 0.$
- $L_n(e_1)(x) = L_n((e_1 - x)^1)(x) = 0.$
- $L_n(e_2)(x) = L_n((e_1 - x)^2)(x) = 3x(x - 1)/(2n(2n - 1)) = O(1/n^2).$
- $L_n(e_3)(x) = n^3 x^2(2nx - 4x + 3)/(2n - 1).$
- $LA_n((e_1 - x)^2)(x) = -x(3n - 2) \cdot (x - 1)/(n(2n - 1)) = O(1/n).$
- $LB_n((e_1 - x)^2)(x) = -x(6n - 1) \cdot (x - 1)/(2n(2n - 1)) = O(1/n).$
- $(LA_n + LB_n)((e_1 - x)^2)(x) = LA_n(\text{abs}(e_1 - x)^2)(x) + LB_n(\text{abs}(e_1 - x)^2)(x) = -x(12n - 5) \cdot (x - 1)/(2n(2n - 1)) = O(1/n).$

To find values like those just listed, it is useful to calculate raw moments (Wang et al. 2023)⁴⁵ and central moments (Weisstein)⁴⁶ of hypergeometric random variables (such as X_k). Indeed, if $g(y) = W_{2n}(e_r; k/(2n)) - W_n(e_r; y)$ is a polynomial in y of degree r or less, then $L_n(e_r)$ can be found using a Taylor expansion, namely as—

$$\begin{aligned} L_n(e_r) &= \sum_{i=0}^r \mathbb{E}[(X_k/n - \mathbb{E}[X_k/n])^i] \frac{g^{(i)}(\mathbb{E}[X_k/n])}{i!} \\ &= \sum_{i=0}^r \frac{\mathbb{E}[(X_k - \mathbb{E}[X_k])^i]}{n^i} \frac{g^{(i)}(k/(2n))}{i!}, \end{aligned}$$

where the derivatives are taken with respect to y , and where $\mathbb{E}[(X_k - \mathbb{E}[X_k])^i]$ is the i -th central moment of X_k .

In the following, the notation $\|f\|$ means $\max_{0 \leq \lambda \leq 1}(\text{abs}(f(\lambda)))$.

The first step is to **find the Taylor expansion** of $L_n(f)(\lambda)$. Given that $L_n((e_1 - x)^0)(x) = L_n((e_1 - x)^1)(x) = 0$, this becomes:

$$L_n(f)(\lambda) = L_n(R_3(f, \lambda)) + \sum_{i=2}^3 L_n((e_1 - \lambda)^i)(\lambda) \frac{f^{(i)}(\lambda)}{i!},$$

$$\text{abs}(L_n(f)(\lambda)) \leq \|L_n(R_3(f, \lambda))\| + \|L_n((e_1 - \lambda)^2)\| \|f^{(2)}\|/2$$

$$+ \|L_n((e_1 - \lambda)^3)\| \|f^{(3)}\|/6.$$

⁴⁵Wang, Y.Q., Zhang, Y.Y, Liu, J.L., “Expectation identity of the hypergeometric distribution and its application in the calculations of high-order origin moments”, Communications in Statistics-Theory and Methods 52(17), 2023. <https://doi.org/10.1080/03610926.2021.2024235>

⁴⁶Weisstein, Eric W. “Central Moment.” From MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/CentralMoment.html>

The function $\text{abs}(L_n((e_1 - x)^3)(x))$ has its maximum at $x = 1/2 - \sqrt{3}/6$; and $\text{abs}(L_n((e_1 - x)^2)(x))$ has its maximum at $x = 1/2$, so:

$$\begin{aligned} \text{abs}(L_n(f)(\lambda)) &\leq \|L_n(R_3(f, \lambda))\| + \text{abs}\left(\frac{3\lambda(\lambda - 1)}{2n(2n - 1)}\right)\|f^{(2)}\|/2 \\ &\quad + \|L_n((e_1 - \lambda)^3)\| \|f^{(3)}\|/6 \\ &\leq \|L_n(R_3(f, \lambda))\| + \frac{3}{8n(2n - 1)}\|f^{(2)}\|/2 \\ &\quad + \frac{\sqrt{3}(6n - 5)}{24n^2(2n - 1)}\|f^{(3)}\|/6. \end{aligned}$$

Meanwhile the remainder is estimated as follows, using the proof of corollary 2.3 of Gonska et al. (2006)⁴⁷:

$$\|L_n(R(f, \lambda))\| \leq \frac{1}{6}\|f^{(3)}\| \|(LA_n + LB_n)(\text{abs}(e_1 - \lambda)^3)\|.$$

In turn, using Schwarz's inequality (see proof of the same paper's corollary 2.1):

$$\begin{aligned} \|(LA_n + LB_n)(\text{abs}(e_1 - \lambda)^3)\| &\leq (\|(LA_n + LB_n)((e_1 - \lambda)^4)\|)^{1/2} \\ &\quad \times (\|(LA_n + LB_n)((e_1 - \lambda)^2)\|)^{1/2} \leq \frac{3\sqrt{3}}{8n^{3/2}}. \end{aligned}$$

(The expression in the middle takes its maximum at $\lambda = 1/2$; the right-hand side is an upper bound of that expression for all positive integers n .) Altogether:

$$\begin{aligned} \|L_n(f)\| &\leq \frac{3}{8n(2n - 1)}\frac{1}{2}\|f^{(2)}\| \\ &\quad + \left(\frac{3\sqrt{3}}{8n^{3/2}} + \frac{\sqrt{3}(6n - 5)}{24n^2(2n - 1)}\right)\frac{1}{6}\|f^{(3)}\| = LC_n(f) \\ &\leq 0.1875\frac{\|f^{(2)}\|}{n^{3/2}} + \frac{5\sqrt{3}}{72}\frac{\|f^{(3)}\|}{n^{3/2}} \leq \frac{0.3078M}{n^{3/2}} = O(1/n^{3/2}). \end{aligned}$$

If $n \geq 2$ is an integer, $LC_n(f) \leq 0.2165M/n^{3/2}$.

⁴⁷Gonska, Heiner, Paula Pi ul, and Ioan Raşa. "On differences of positive linear operators." Carpathian Journal of Mathematics (2006): 65-78.

8 Example: An Interesting Linear Operator

For a continuous function f on the closed unit interval and for nonnegative integers m and n , let $H_{n,m}$ be a linear operator as follows:

$$H_{n,m}(f) = B_n(f) + \text{Lag}_m(f) - B_n(\text{Lag}_m(f)),$$

where B_n is the degree- n Bernstein polynomial and Lag_m is the polynomial of degree up to m that equals f at “ $m+1$ distinct points on” the closed unit interval. This operator was mentioned in Remark 2 of Gavrea and Ivan (2018)⁴⁸, but appears not to have been studied elsewhere.

It is known that Lag_m is a linear operator and reproduces all polynomials of degree m or less, so that $\text{Lag}_m(e_i) = e_i$ whenever $0 \leq i \leq m$ is an integer. Thus, if f is such a polynomial, $B_n(f) = B_n(\text{Lag}_m(f))$ and therefore $H_{n,m}(f) = \text{Lag}_m(f) = f$, and therefore $H_{n,m}(e_i) = e_i$ whenever $0 \leq i \leq m$ is an integer.

(The foregoing sentence would remain true if B_n were replaced with any other operator mapping to and from the same functions.)

Because $H_{n,m}$ is linear and reproduces all polynomials up to degree m , the following holds if f has a continuous m -th derivative:

$$\begin{aligned} H_{n,m}(f)(\lambda) - f(\lambda) &= H_{n,m}(R_m(f, \lambda))(\lambda) \\ &= B_n(R_m(f, \lambda)) + \text{Lag}_m(R_m(f, \lambda)) - B_n(\text{Lag}_m(R_m(f, \lambda))). \end{aligned}$$

With the help of Lemma 6, the following holds if n is also 2 or greater:

$$\|H_{n,m}(f)(\lambda) - f(\lambda)\| \leq \frac{\|f^{(m)}\| \mu_r}{(r!)n^{r/2}} + \|\text{Lag}_m(R_m(f, \lambda)) - B_n(\text{Lag}_m(R_m(f, \lambda)))\|,$$

where μ_r is as in Proposition 1 and the notation $\|f\|$ means $\max_{0 \leq \lambda \leq 1} (\text{abs}(f(\lambda)))$.

Alternatively, write:

$$\begin{aligned} H_{n,m}(f) - f &= B_n(f) + \text{Lag}_m(f) - B_n(\text{Lag}_m(f)) - f \\ &= (B_n(f) - f) + (\text{Lag}_m(f) - B_n(\text{Lag}_m(f))), \end{aligned}$$

$$\text{abs}((H_{n,m}(f) - f)(\lambda)) \leq \text{abs}(B_n(f) - f) + \text{abs}(B_n(\text{Lag}_m(f)) - \text{Lag}_m(f)),$$

so now there are two error bounds to find: one for f and the other for $\text{Lag}_m(f)$. And, if f has a continuous second derivative, both have the same form:

$$B_n(g) \leq M_2(g)/(8n),$$

⁴⁸Ioan Gavrea, Mircea Ivan, “A note on the fixed points of positive linear operators”, Journal of Approximation Theory (227), 2018, <https://doi.org/10.1016/j.jat.2017.12.001>.

where $M_i(g)$ is the maximum absolute value of g 's i -th derivative. (This follows from Lorentz (1963)⁴⁹ and the well-known fact that M_2 is an upper bound of g 's first derivative's smallest Lipschitz constant.) Thus what is left is to estimate the second derivative of $\text{Lag}_m(f)$. Given that that function is a polynomial of degree m or less, this can be estimated as:

$$\text{abs}(\text{Lag}_m(f)^{(2)}(\lambda)) \leq \|\text{Lag}_m\| M_0(f) \cdot \max(1, m)^2,$$

where $\|\text{Lag}_m\|$ is the operator norm of Lag_m , also known as its *Lebesgue constant*, which will vary depending on the points on the closed unit interval where the polynomial meets (interpolates) f . The inequality just shown relies on Bernstein's inequality for the derivatives of polynomials (Weisstein)⁵⁰.

Altogether, if f has a continuous second derivative and m is fixed:

$$\text{abs}((H_{n,m}(f) - f)(\lambda)) \leq \frac{M_2(f)}{8n} + \frac{\|\text{Lag}_m\| M_0(f) \cdot \max(1, m)^2}{8n}.$$

Example: If m is 3, and the polynomial generated by Lag_m interpolates f at the points 0, 1/3, 2/3, and 1, the inequality just shown becomes:

$$\text{abs}((H_{n,3}(f) - f)(\lambda)) \leq \frac{M_2(f)}{8n} + \frac{1.64 \cdot M_0(f) \cdot 9}{8n},$$

using an upper bound for $\|\text{Lag}_3\|$.

9 Probabilistic Interpretations of Linear Operators

The Bernstein polynomials featured in a proof in 1912 of the result that any continuous function on a closed interval can be approximated as well as desired by polynomials (Bernstein 1912)⁵¹. That proof used probability theory. In a series of papers, Adell and De la Cal use probability theory to interpret a number of linear operators in addition to those polynomials (Adell and De la Cal 1996⁵², 1995⁵³).

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11 Notes

⁴⁹G.G. Lorentz, "Inequalities and saturation classes for Bernstein polynomials", 1963.

⁵⁰Weisstein, Eric W. "Bernstein's Inequality." From MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/BernsteinsInequality.html>

⁵¹S.N. Bernstein, "Démonstration du théorème de Weierstrass fondée sur le calcul de probabilités", Comm. Kharkov Math. Soc. 13, 1-2, 1912.

⁵²Adell, J. A., and J. De la Cal. "Bernstein-type operators diminish the ϵ -variation." Constructive Approximation 12.4 (1996): 489-507. <https://doi.org/10.1007/BF02437505>

⁵³Adell, J. A., and J. De la Cal. "Bernstein-Durrmeyer operators." Computers & Mathematics with Applications 30.3-6 (1995): 1-14. [https://doi.org/10.1016/0898-1221\(95\)00081-X](https://doi.org/10.1016/0898-1221(95)00081-X)

⁵⁴<https://creativecommons.org/publicdomain/zero/1.0/>